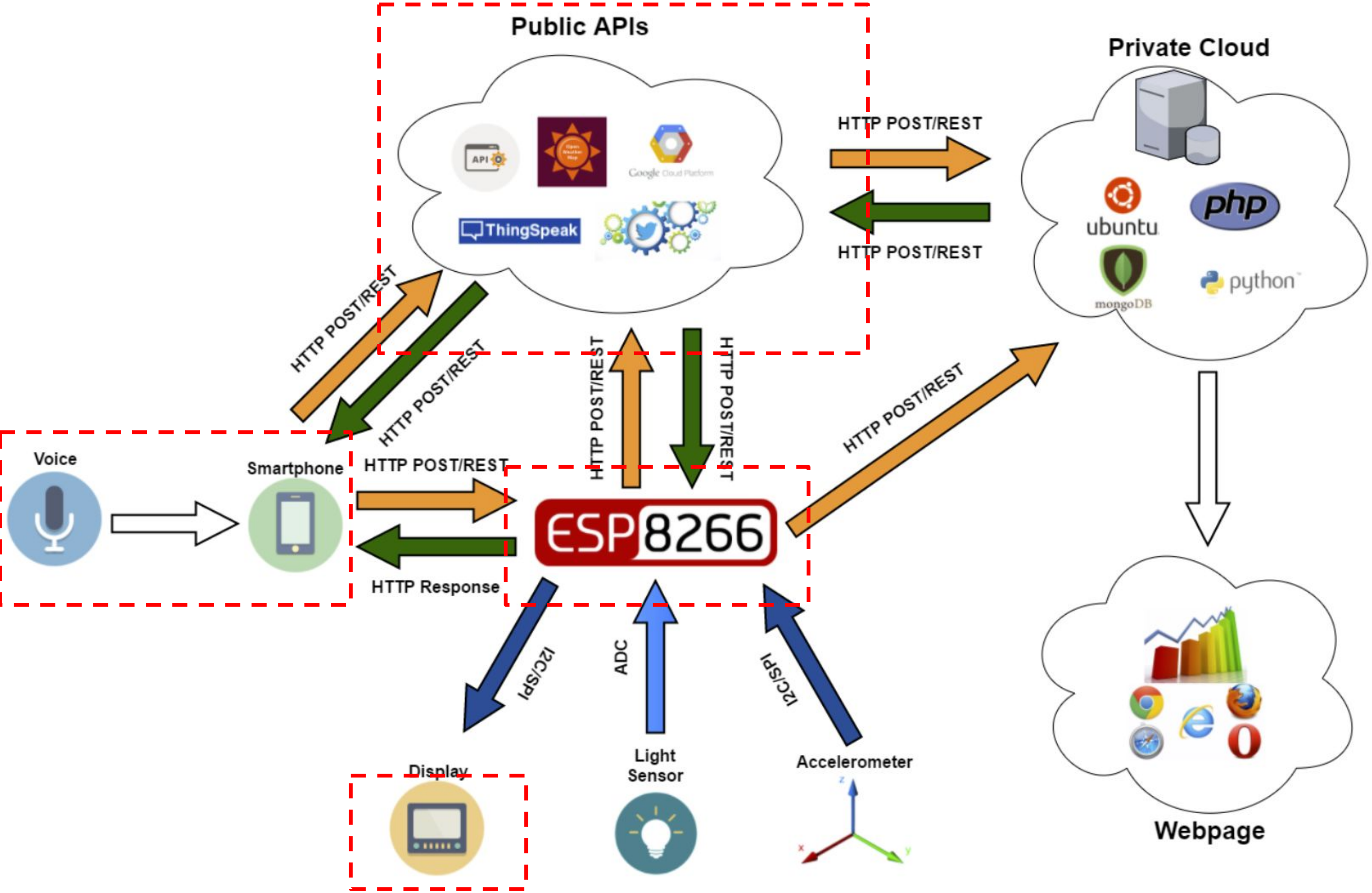
IoT - Intelligent and Connected Systems
EECS E4764 Fall' 22

Lab 5

October 12th, 2022

Lab Overview



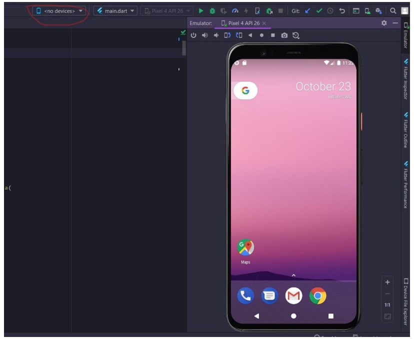
Part 1

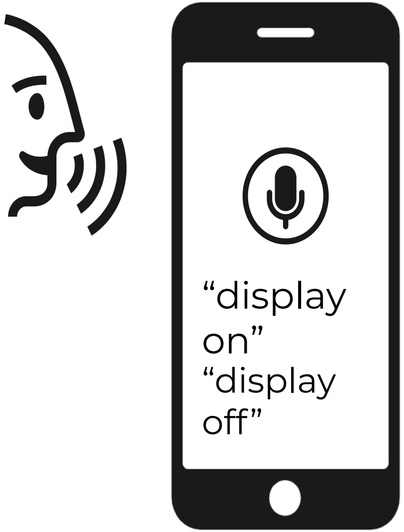
Voice to text using smartphone

Part 1

- Simple smartphone App
- Use your favorite framework
- No experience? No problem.
    - Many examples and tutorials on the Internet
    - Good starting point: **Android Studio** for Android apps
        - Includes emulator (so you don’t need an Android phone)
        - Language: Java
- Make sure to configure permissions

(described in lab manual)

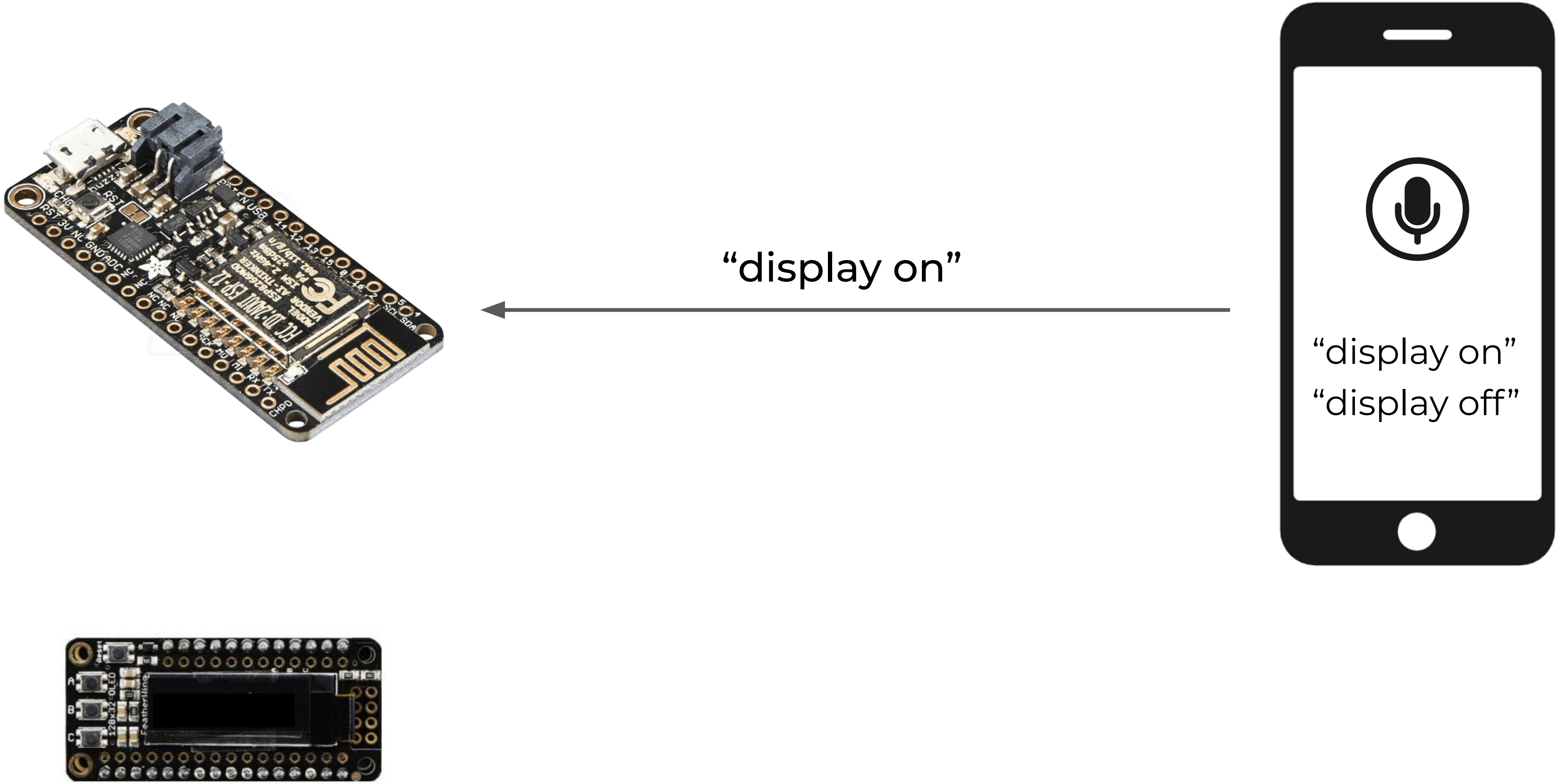




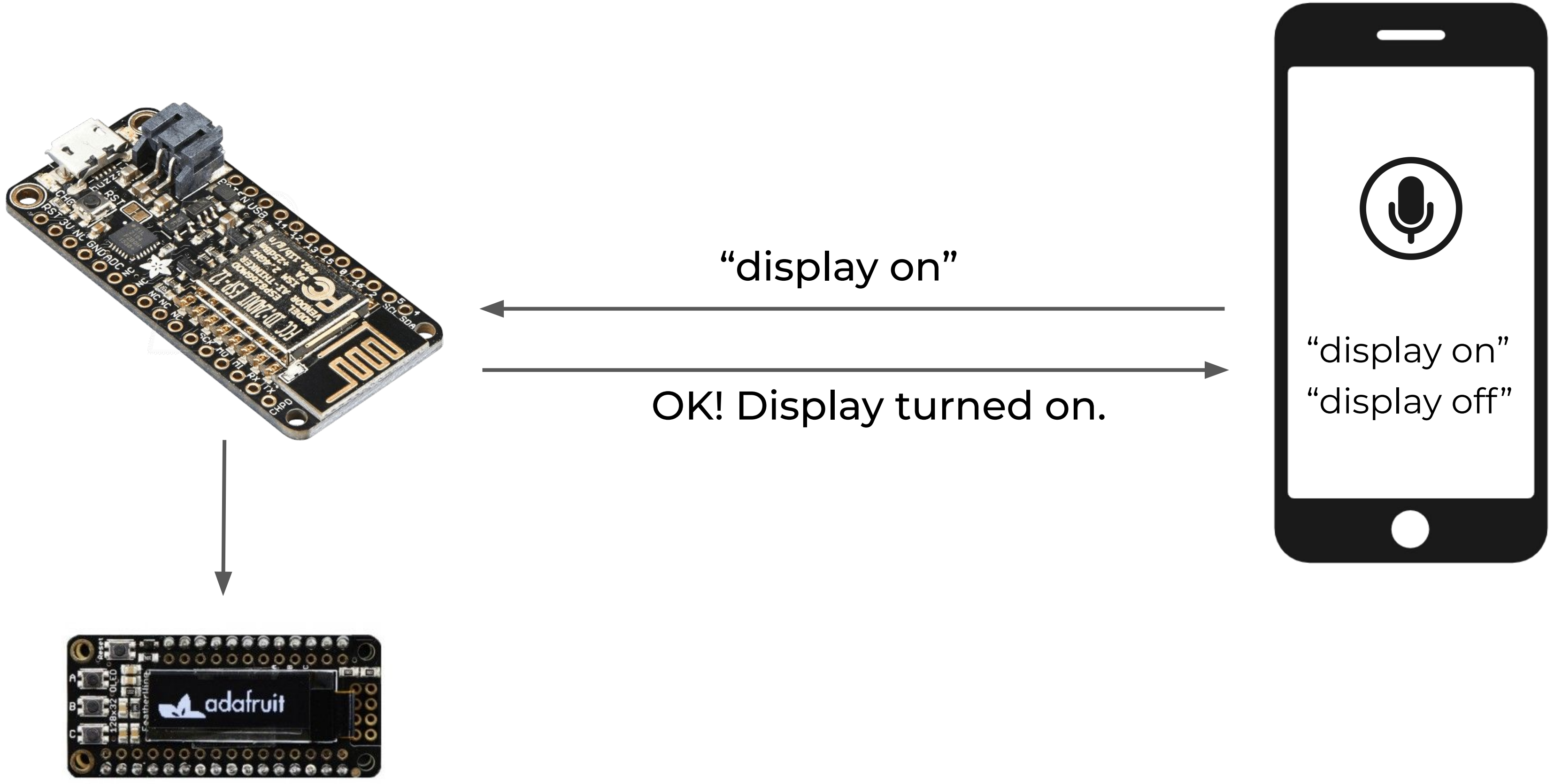
Part 2

Embedded Server on Huzzah

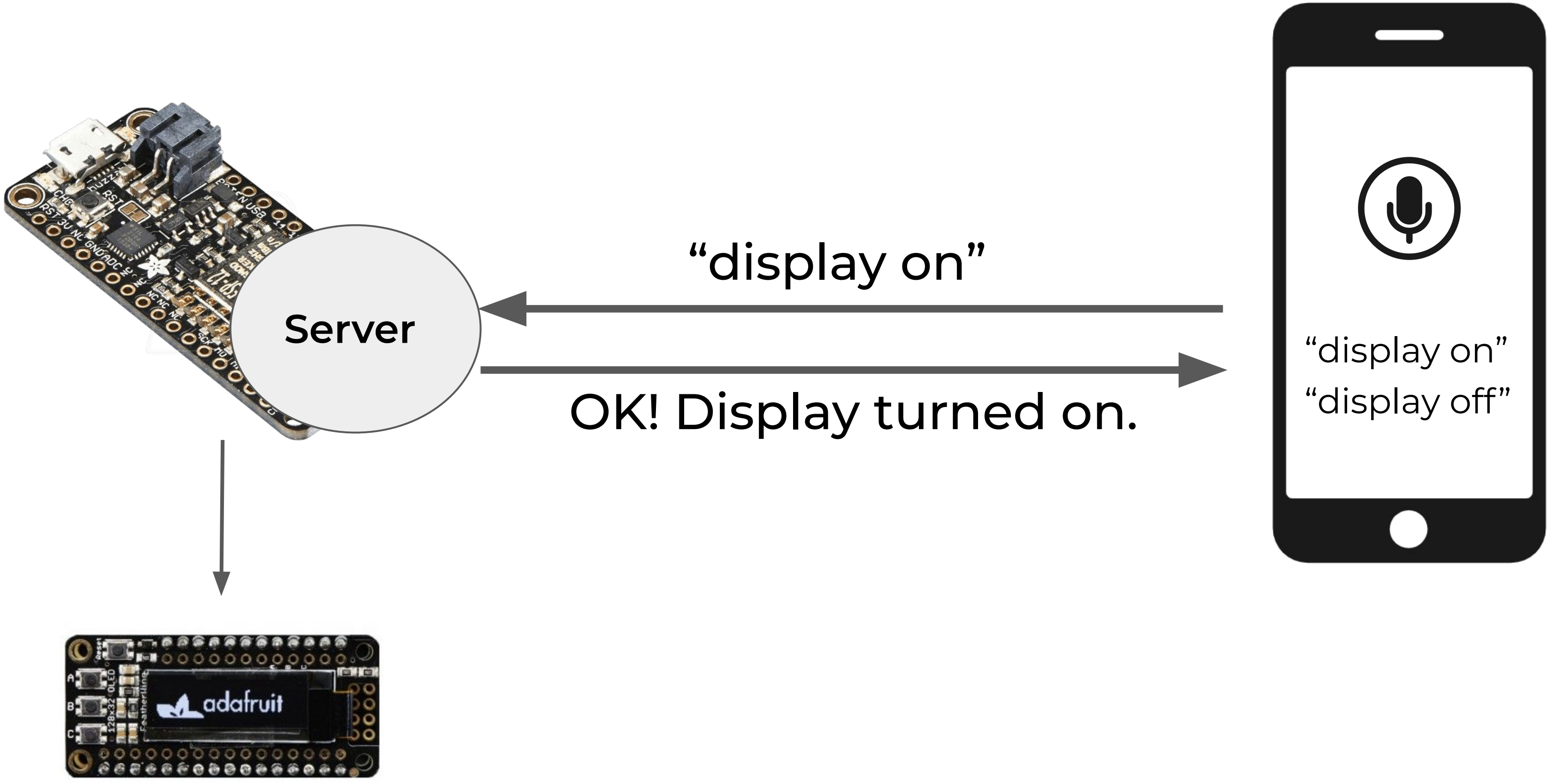
Part 2 Overview



Part 2 Overview



How to listen to commands and respond?



How to build a simple HTTP server?

- Use socket in MicroPython!
- Usage almost same as on computer Python

```python
addr = ('0.0.0.0',80)
s = socket.socket()
s.bind(addr)
s.listen(1)
while(1):
    try:
        (conn, address) = s.accept()
    except OSError:
        print("Nothing")
    else:
        rec = conn.recv(4096)
        rec = rec.split(b'\r\n\r\n')
        
```

https://docs.micropython.org/en/latest/library/socket.html
https://docs.micropython.org/en/latest/library/socket.html#socket.socket.settimeout
https://docs.micropython.org/en/latest/library/socket.html#socket.socket.setblocking

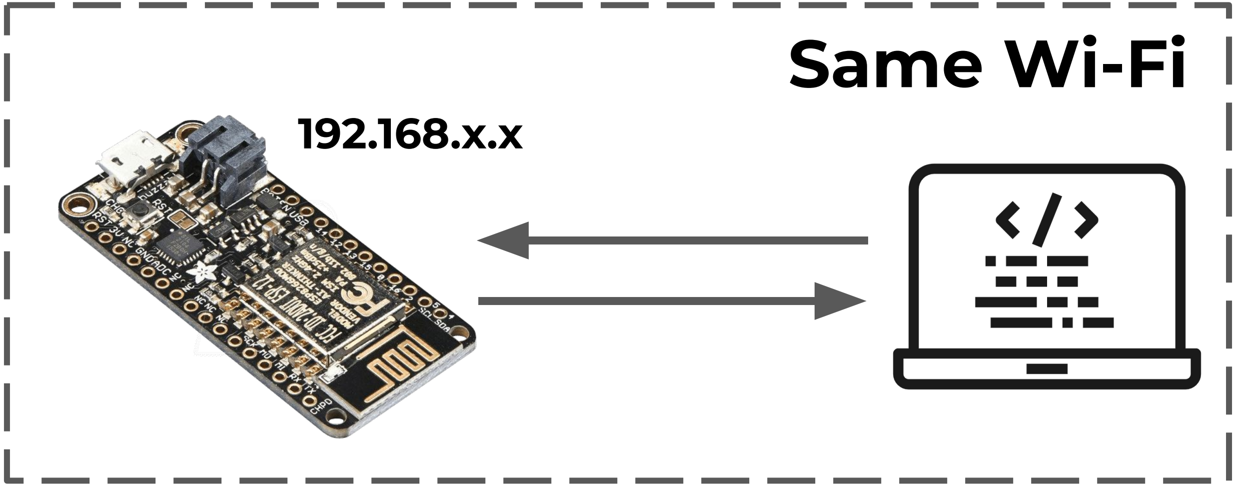
Testing

- Develop and test code in Python first, before putting into ESP
- Test with a computer client running Python

Testing

- Develop and test code in Python first, before putting into ESP
- Test with a computer client running Python
  - Use ESP's Local IP Address

```
wlan = network.WLAN(network.STA_IF)
...
# After wlan.isconnected()
print(wlan.ifconfig())
```

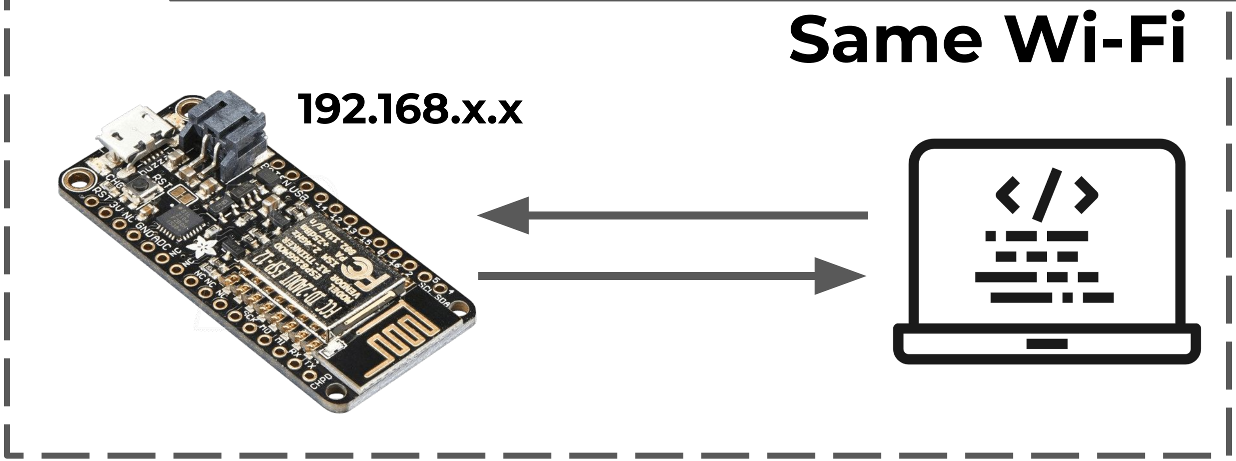


Testing

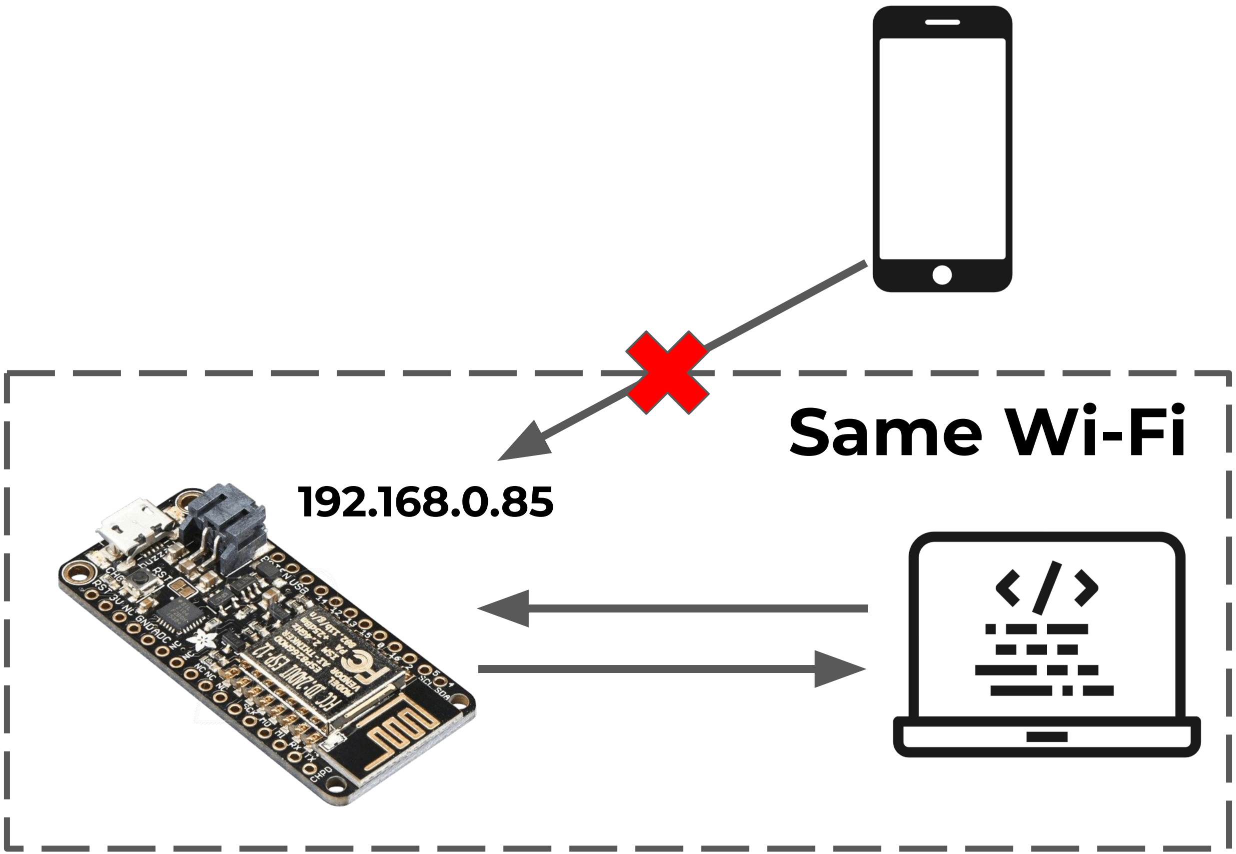
- Develop and test code in Python first, before putting into ESP
- Test with a computer client running Python
    - Use ESP's Local IP Address

```
wlan = network.WLAN(network.STA_IF)
...
```

Will this IP address work on our phone?

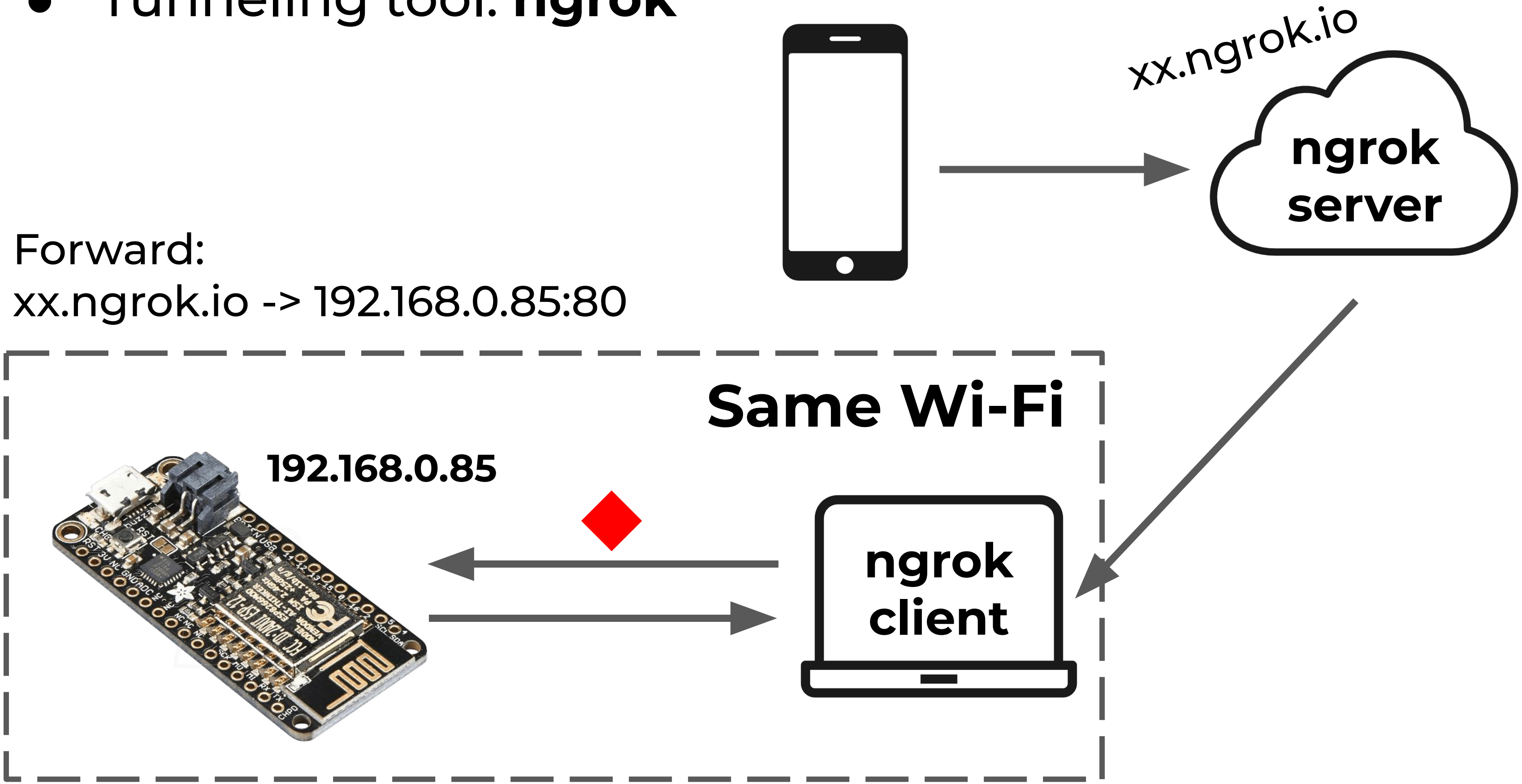


Not always!



How to access from our phone?

- Tunneling tool: **ngrok**



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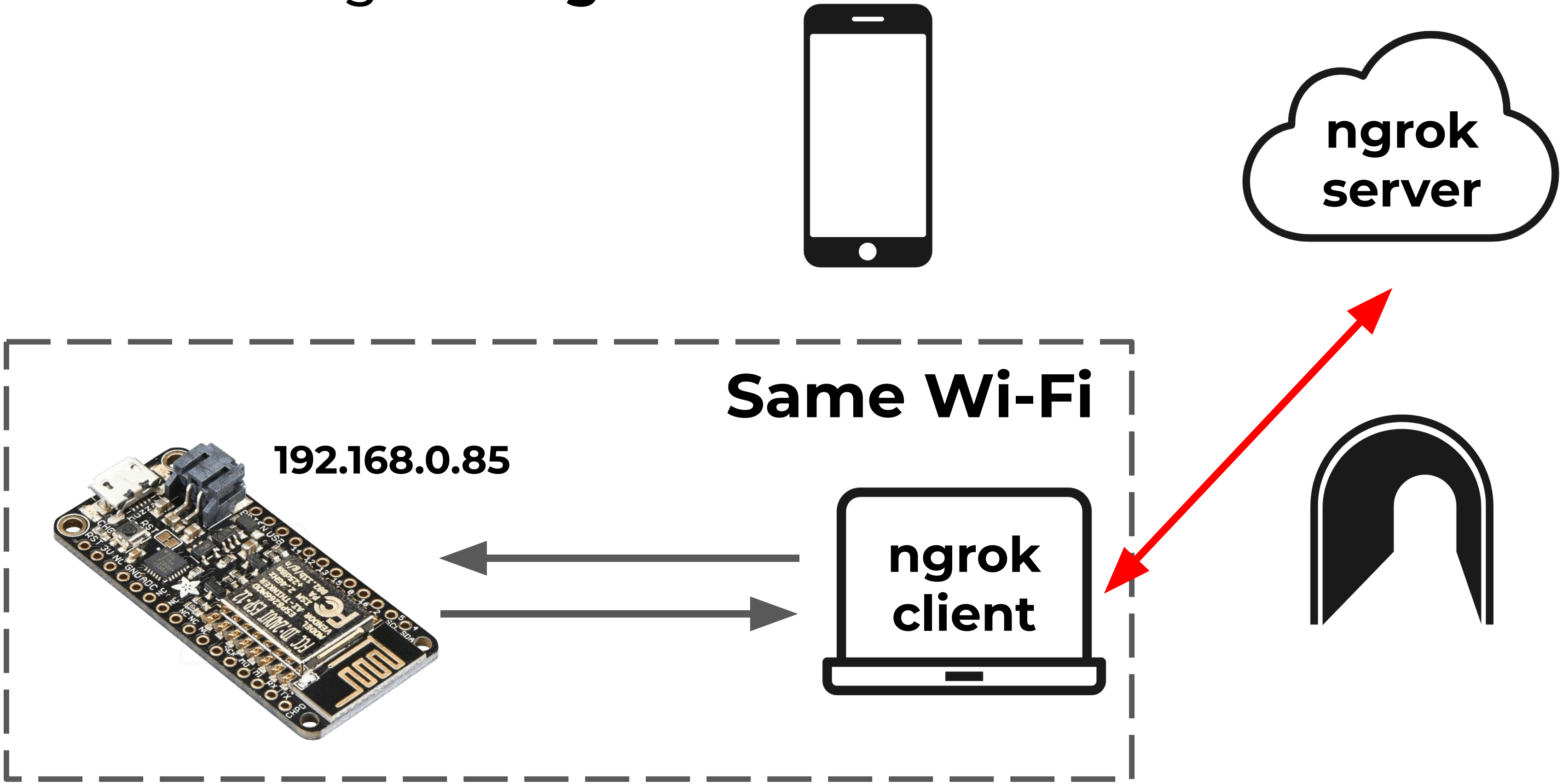

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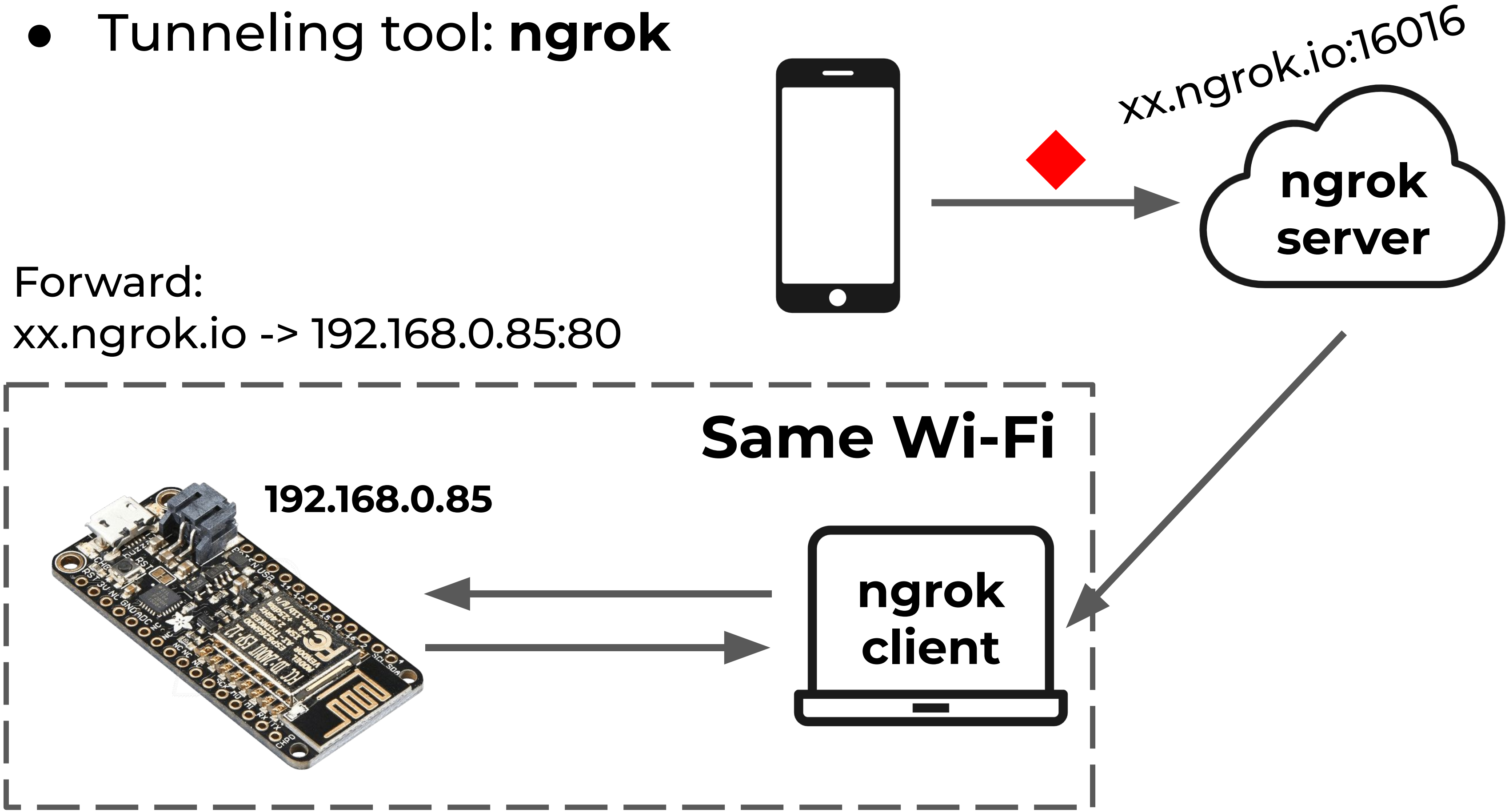

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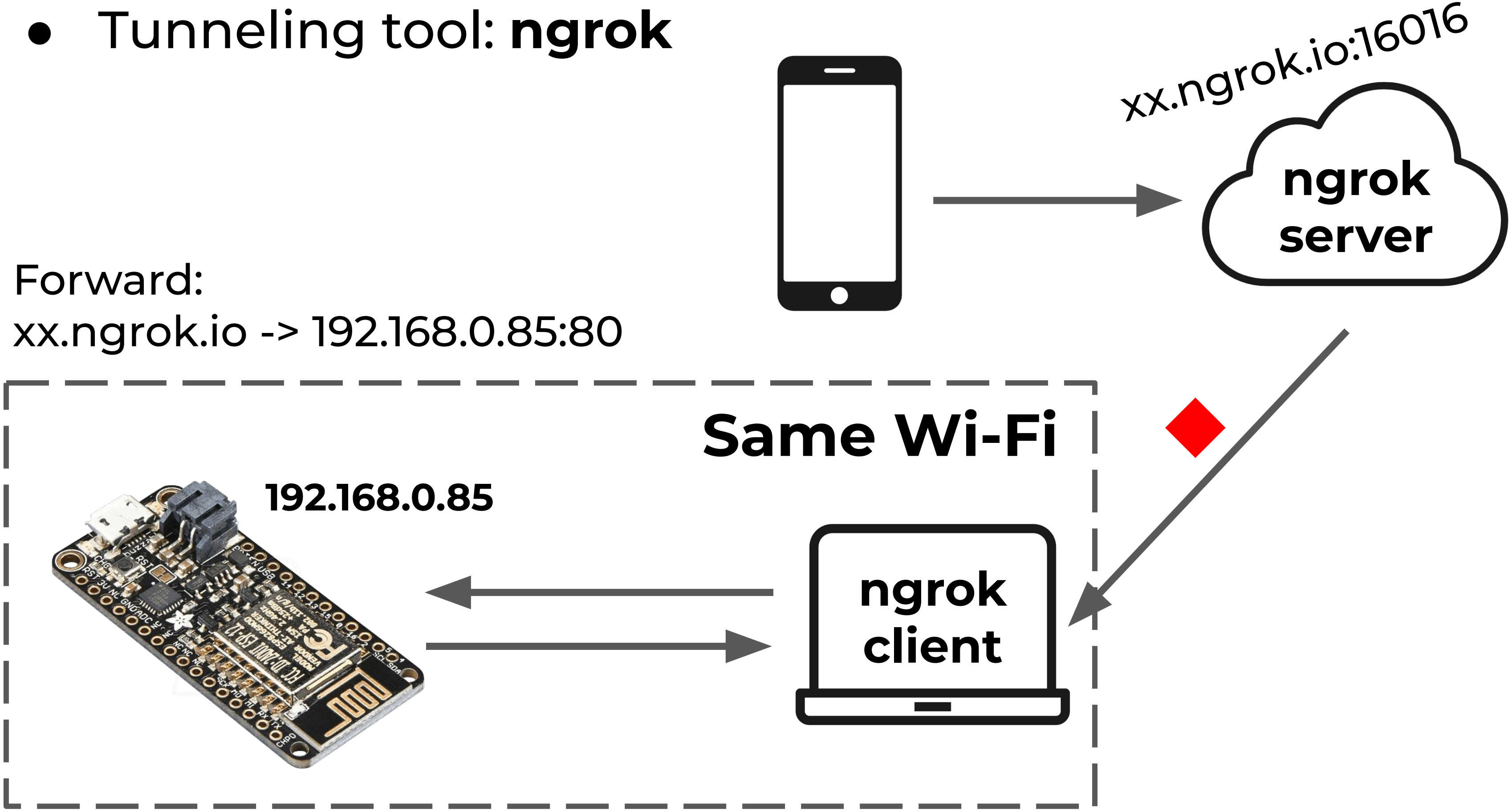

How to access from our phone?

https://ngrok.com/docs

```
ngrok <http/tcp> <ESP_IP_ADDRESS>:<PORT>
```

```
ngrok                                                                   ⌥⌘1
ngrok by @inconshreveable                                    (Ctrl+C to quit)

Session Status                online
Account                       Scott Zhao (Plan: Free)
Version                       2.3.40
Region                        United States (us)
Web Interface                 http://127.0.0.1:4040
Forwarding                    http://4cb7-160-39-153-136.ngrok.io -> http://192.168.0.85:80
Forwarding                    https://4cb7-160-39-153-136.ngrok.io -> http://192.168.0.85:80
```

What to expect

- Implement commands:
  - “display on/off”: turn on or off the OLED display
  - “display time”: behave like a regular digital watch
  - Other messages: display the message on the OLED display
- Two-way communication
  - ESP sends back a response of command execution status
  - App displays the confirmation message received

Lab 5 Checkpoints

1. Make a simple application that enables voice recognition. Display the text on the screen after speaking to the microphone.
2. Use the voice commands to turn off/on the display, display the time, and display any other spoken message on the OLED screen.

Logistics

- Due in 2 weeks — Wednesday, October 26th
- Many components can be developed and tested in parallel -- team work!
- Next week: Project idea feedback session

  → Have project ideas ready to pitch!